

Blinded Inverse Takeoff

Finite-horizon limits in recovering recursive redundancy

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Abstract. Memoization collapses repeated calls in a recursion tree into shared states. The velocity takeoff profile records how quickly that redundant work appears as input size grows. This note studies the inverse question: can a finite, noisy takeoff trace reveal the recurrence that produced it? Our contributions are fivefold. We (i) characterize exactly what a finite observed prefix determines; (ii) construct exponentially large fixed-invariant ambiguity classes, obtain a minimax lower bound under arbitrary parameter-independent noise, and show that finite holdout cannot certify mechanism; (iii) prove that for known maximum lag D , exactly $D - 1$ samples suffice uniformly, whereas $D - 2$ do not, and give a constructive decoder; (iv) prove that fixed speed and aperiodicity still permit arbitrarily slow settling; and (v) test a generic matrix-pencil baseline and constraint-aware source decoder in three blinded AWS sweeps containing 1.20 million hidden schemas and 35.0 million trials. The baseline succeeds on many visible short-lag recurrences but fails sharply on the hidden-lag family. Under noise, accurate forecasting often survives after modal identification collapses. Prediction, modal explanation, and recursive source identification are distinct claims.

1 Why invert a takeoff curve?

A naive recursive program exposes a tree of calls. Equal labels can appear many times; memoization identifies them and leaves a directed acyclic graph (DAG) of distinct subproblems. For input size $n \geq 0$, let N_n be the number of calls in the naive tree and let M_n be the number of distinct reachable labels. Throughout, $\log$ denotes the natural logarithm. The logarithmic overlap is

$$S_n = \log \frac{N_n}{M_n},$$

with one-step velocity

$$v_n = S_{n+1} - S_n = u_n - w_n,$$

where

$$u_n = \log N_{n+1} - \log N_n, \quad w_n = \log M_{n+1} - \log M_n.$$

If N_n grows exponentially and $\alpha := \lim_{n \rightarrow \infty} u_n > 0$ exists while M_n is polynomial, then $w_n \rightarrow 0$ and $v_n \rightarrow \alpha$. That terminal number says how fast redundancy eventually accumulates, but not how it gets there. Two recurrences can have the same cruising speed while one accelerates immediately and the other rings for a very long time.

The normalized takeoff profile and its causal difference are

$$F_n = \frac{u_n}{\alpha}, \quad \kappa_n = F_n - F_{n-1}, \quad F_{-1} = 0.$$

Thus $F_n \rightarrow 1$, and κ records the response shape: delay, monotone settling, alternating correction, or damped ringing. Forward analysis starts from a recurrence and predicts F . Inverse takeoff starts from an observed prefix of F and asks what can be recovered.

A lag $j \geq 1$ is measured in input steps. An *observation horizon* $T \geq 1$ means that exactly the T samples $F_0, \dots, F_{T-1}$ are available. Thus “length T ” counts samples rather than the largest displayed index. A *forecast horizon* $h \geq 1$ refers to the next h samples $F_T, \dots, F_{T+h-1}$. When a maximum admissible lag is assumed, $D \geq 1$ denotes that known bound.

The analogy is system identification, but the object is unusual. The “plant” is the redundant response of a recursion tree before memoization quotients equal states. Classical Prony and matrix-pencil methods can estimate damped exponential modes from samples [1, 2, 3]. What is new is the constrained source class and the distinction between response modes and literal recursive branches.

Contributions. The note makes five concrete advances. **First**, an exact difference identity and root certificate lead to a complete characterization: a profile’s first T samples reveal total branching and precisely the coefficients at lags below T . **Second**, the unseen tail can contain exponentially many aperiodic integer recurrences with the same speed, branch count, DAG-growth class, and observed prefix; the resulting source ambiguity survives arbitrary parameter-independent noise and any finite holdout placed before the hidden lag returns. **Third**, the negative result has a sharp constructive boundary: given a lag bound D , the $D - 1$ exact samples through F_{D-2} recover every coefficient, while $D - 2$ samples are insufficient in general. **Fourth**, a fixed-speed family proves that aperiodicity does not bound the settling time, explaining the long echo trains seen in the kernels. **Fifth**, 35.0 million blinded trials map where a classical spectral estimator works, where it fails, and show a constraint-aware decoder attaining the exact boundary while accurate forecasts can coexist with a wrong source.

2 Finite-lag response and its modes

Consider the finite-lag schema

$$N_n = 1 + \sum_{j \in J} a_j N_{n-j}, \quad N_q = 1 \text{ for } q \leq 0, \quad (1)$$

where $J \subset \{1, 2, \dots\}$ is finite and $a_j \in \mathbb{Z}_{>0}$ (the positive integers) is the number of branches at lag j . The symbol q is only a boundary index. For a complex variable z , define the recurrence polynomial

$$Q(z) = 1 - \sum_{j \in J} a_j z^j.$$

We call the schema *aperiodic* when the greatest common divisor of the active lag set J is one. When $\gcd(J) = 1$ and $\sum_j a_j > 1$, Q has a unique smallest-modulus positive root $\rho \in (0, 1)$, and the terminal speed is $\alpha = -\log \rho$. Non-dominant singularities ζ contribute normalized correction modes

$$\theta = \frac{\rho}{\zeta}.$$

The affine $+1$ in (1) also contributes the singularity $z = 1$. The largest active $|\theta| < 1$ controls the eventual settling rate, while a complex leading mode produces ringing.

For a common terminal speed $\alpha = \log 2$, the root is $\rho = 1/2$ and the coefficient constraint becomes

$$\sum_j a_j 2^{-j} = 1. \quad (2)$$

This dyadic identity is useful experimentally: it generates many structurally different recurrences while holding the first-order asymptotic invariant exactly fixed.

An observed tail is modeled approximately as

$$Y_n := F_n - 1 \approx \sum_{r=1}^R c_r \theta_r^n. \quad (3)$$

Here Y_n is the deviation from terminal takeoff, R is the retained mode count, c_r is the amplitude of mode r , and θ_r is its normalized pole. Complex modes and amplitudes occur in conjugate pairs because Y_n is real. The approximation matters. Even when N_n has a finite exponential expansion, the logarithmic ratio defining F_n introduces higher-order mode products. A low-rank spectral fit is therefore a tail model, not an algebraic identity for every finite prefix.

Figure 1 gives a visual vocabulary for the inverse problem. The top row is the takeoff profile; the bottom row is its causal difference, the kernel actually carrying the transient shape. The same terminal level can be approached smoothly, by sign alternation, through damped oscillation, or in widely separated echoes.

3 What a finite trace can contain

The obstruction is more basic than numerical noise: a recursive signal cannot reveal a branch whose lag has not returned. The following sequence makes that idea exact and also gives its positive converse. Extend every schema by $a_j = 0$ off its finite support, and write $A = \sum_j a_j$.

Lemma 1 (Exact difference response). *Let $d_n = N_n - N_{n-1}$ for $n \geq 1$ and $d_n = 0$ for $n \leq 0$. Then*

$$d_1 = A, \quad d_n = \sum_j a_j d_{n-j} \quad (n \geq 2), \quad (4)$$

and, as a formal power series,

$$G_d(z) := \sum_{n \geq 1} d_n z^n = \frac{Az}{Q(z)}.$$

Proof. The boundary convention gives $N_1 = 1 + A$, hence $d_1 = A$. Subtracting (1) at $n - 1$ from the equation at n gives (4); multiplying by z^n and summing gives the quotient. $\square$

Lemma 2 (Fixed-speed root certificate). *Suppose $a_j \geq 0$, (2) holds, and $\gcd\{j : a_j > 0\} = 1$, where $\gcd$ denotes greatest common divisor. Then $\rho = 1/2$ is the unique root of Q in $|z| \leq 1/2$. Consequently the terminal speed is $\alpha = \log 2$.*

Proof. If $Q(z) = 0$ and $|z| \leq 1/2$, then

$$1 = \left| \sum_j a_j z^j \right| \leq \sum_j a_j |z|^j \leq \sum_j a_j 2^{-j} = 1.$$

Equality forces $|z| = 1/2$ and all active phases $z^j/|z|^j$ to equal one. Their lag $\gcd$ is one, so $z = 1/2$. $\square$

Proposition 3 (No uniform settling rate). *For integers $L \geq 2$, consider the aperiodic fixed-speed family*

$$Q_L(z) = 1 - (2^L - 1)z^L - 2z^{L+1},$$

and define

$$r_L := \max_{\substack{Q_L(\zeta)=0 \\ \zeta \neq 1/2}} \left| \frac{1/2}{\zeta} \right|.$$

Then $r_L \rightarrow 1$ as $L \rightarrow \infty$. Thus fixed terminal speed and aperiodicity do not impose a uniform transient length.

Proof. Put $x = 2z$. The root equation becomes $f_L(x) + g_L(x) = 0$, where $f_L = 1 - x^L$ and $g_L = 2^{-L}x^L(1 - x)$. On a circle of radius $C2^{-L}/L^2$ about $\omega_L = e^{2\pi i/L}$, Taylor expansion gives $|f_L| \sim C2^{-L}/L$ and $|g_L| \sim 2\pi 2^{-L}/L$. Choosing $C > 2\pi$, Rouché's theorem gives a root $x_L = \omega_L + O(2^{-L}/L^2)$. Lemma 2 gives $|x_L| > 1$, while $|x_L| \rightarrow 1$; hence the associated normalized radius $|1/x_L| \rightarrow 1$. $\square$

Theorem 4 (Complete finite-prefix characterization). *Consider schemas with coefficient sequences $a = (a_j)$ and $b = (b_j)$ and a common known terminal speed α . Let $A_a = \sum_j a_j$ and $A_b = \sum_j b_j$. For every observation horizon $T \geq 1$, their profiles agree on $F_0, \dots, F_{T-1}$ if and only if $A_a = A_b$ and*

$$a_j = b_j \quad (1 \leq j < T).$$

Proof. The prefix reconstructs

$$N_n = \exp\left(\alpha \sum_{r=0}^{n-1} F_r\right), \quad 1 \leq n \leq T,$$

and hence $d_1, \dots, d_T$. Lemma 1 gives $A = d_1$ and the triangular inversion

$$a_k = \frac{d_{k+1} - \sum_{j=1}^{k-1} a_j d_{k+1-j}}{d_1}, \quad 1 \leq k < T. \quad (5)$$

Thus equality of profiles implies the stated equalities. Conversely, (4) shows inductively that those equalities force $d_1, \dots, d_T$, and therefore the profile prefixes, to agree. $\square$

Theorem 5 (Sharp recovery boundary with a lag bound). *Let the known integer $D \geq 3$ be an upper bound on the largest lag, so that $a_j = 0$ for $j > D$, and let the terminal speed $\alpha > 0$ be known. The $T = D - 1$ noiseless samples $F_0, \dots, F_{D-2}$ always give exact recovery. For the fixed speed $\alpha = \log 2$, no smaller T works uniformly; hence the sample threshold is sharp.*

Proof. Put $\rho = e^{-\alpha} \in (0, 1)$. Theorem 4 recovers $A = \sum_j a_j$ and $a_1, \dots, a_{D-2}$. Define the remaining branch count and root mass

$$S = A - \sum_{j=1}^{D-2} a_j, \quad R = 1 - \sum_{j=1}^{D-2} a_j \rho^j.$$

Then $x = a_{D-1}$ and $y = a_D$ are the unique solution of $x + y = S$ and $\rho^{D-1}x + \rho^D y = R$; explicitly,

$$x = \frac{R - \rho^D S}{\rho^{D-1}(1 - \rho)}, \quad y = S - x.$$

Thus $T = D - 1$ suffices. For necessity, set $L = D - 2$ in Proposition 6: its distinct schemas all have maximum lag D and agree through F_{D-3} , the prefix of length $D - 2$. $\square$

Proposition 6 (Exponential fixed-invariant ambiguity). *Fix an observation horizon T and choose an integer base lag $L \geq T$. There are*

$$K_L = \left\lfloor \frac{2^{L+1} - 1}{3} \right\rfloor$$

distinct aperiodic integer schemas with common speed $\log 2$, total branch count 2^{L+1} , and at-most-linear memoized-state growth $M_n = O(n)$, all agreeing on $F_0, \dots, F_{T-1}$.

Proof. For $1 \leq t < 2^{L+1}/3$, set

$$a_L = t, \quad a_{L+1} = 2^{L+1} - 3t, \quad a_{L+2} = 2t.$$

The coefficients are positive, sum to 2^{L+1} , and satisfy $4a_L + 2a_{L+1} + a_{L+2} = 2^{L+2}$, which is (2). Consecutive support gives aperiodicity, so Lemma 2 fixes the speed. Reachable labels are one-dimensional, giving $M_n = O(n)$. Theorem 4 gives the common prefix. Distinct t first separate at F_L and define distinct Q_t . $\square$

Corollary 7 (Noisy minimax obstruction). *Choose uniformly from the K_L schemas in Proposition 6 and pass the common prefix through any parameter-independent noise channel, meaning that the channel law depends on the clean prefix but not on its schema label. Every source estimator $\hat{t}$ satisfies*

$$\Pr(\hat{t} = t) \leq K_L^{-1}, \quad \inf_{\hat{t}} \sup_t \Pr(\hat{t} \neq t) \geq 1 - K_L^{-1}.$$

Here t indexes the true schema, $\hat{t}$ is its estimate, $\Pr$ is the joint probability over the uniform source and noise, and $\Pr_t$ conditions on schema t .

Proof. All K_L observation laws are identical. Under the uniform prior, the posterior remains uniform, giving the first bound; some parameter has success at most the average, giving the minimax bound. $\square$

Corollary 8 (Finite holdout does not certify mechanism). *For arbitrary observation horizon T and forecast horizon h , distinct fixed-invariant schemas can agree on both the observed samples $F_0, \dots, F_{T-1}$ and the forecast targets $F_T, \dots, F_{T+h-1}$ while having different recurrence polynomials and full modal spectra.*

Proof. Use Proposition 6 with $L \geq T + h$. The profiles agree through F_{T+h-1} , while the distinct polynomials Q_t , each with constant coefficient one, cannot have the same complete root multiset. $\square$

The picture is now sharp. Without a lag bound, the ambiguity class can grow exponentially and survives arbitrary parameter-independent noise. With a lag bound, the sharp threshold is $D - 1$ samples: triangular inversion recovers the visible coefficients and the branch-count and root equations recover the last two. Intuitively, this is listening for an echo before sound reaches the wall: unlimited estimator sophistication cannot recover absent information. Figure 2 displays one such equivalence class and its eventual separation.

4 The blinded experiment

The study used five fixed-speed families: immediate binary recursion; the near-periodic slow family $a_L = 2^L - 1$, $a_{L+1} = 2$, where L is its base lag; the exponentially ambiguous finite-horizon family; a two-scale family with one local and one delayed channel; and random dyadic refinements that replace one branch at lag j by two branches at lag $j + 1$. Every such refinement preserves (2).

For each hidden schema we computed an exact integer call-count sequence and formed F_n . Let $\mathcal{W}$ denote the current set of T observed indices and define

$$\text{RMS}_{\mathcal{W}}(Y) := \left(T^{-1} \sum_{n \in \mathcal{W}} Y_n^2 \right)^{1/2}.$$

The estimator received only

$$\tilde{Y}_n = Y_n + \sigma \max\{\text{RMS}_{\mathcal{W}}(Y), 10^{-12}\} \xi_n, \quad \xi_n \sim \mathcal{N}(0, 1),$$

not the coefficients or true roots. Here $\sigma \geq 0$ is the relative noise level and the ξ_n are independent standard-normal draws. We tested windows beginning at the origin and immediately after the maximum lag $D := \max J$. Relative noise levels were $0, 10^{-8}, 10^{-5}, 10^{-3}$.

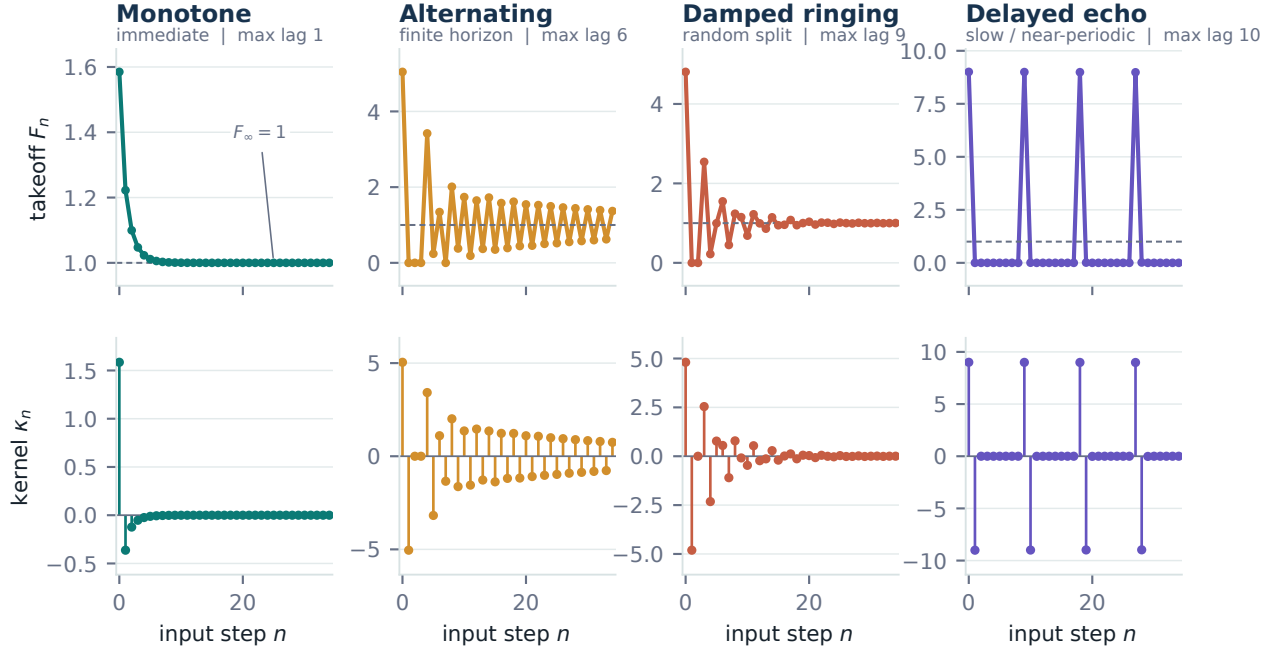


Figure 1: A kernel atlas from representative experimental schemas. All four recurrences have terminal speed $\alpha = \log 2$ and hence $F_n \rightarrow 1$, but their finite responses are plainly different. The dashed line marks the common terminal level. The lower row shows $\kappa_n = F_n - F_{n-1}$: a short relaxation, alternating correction, damped ringing, and a near-periodic echo train. Different vertical scales are used so that each transient remains legible.

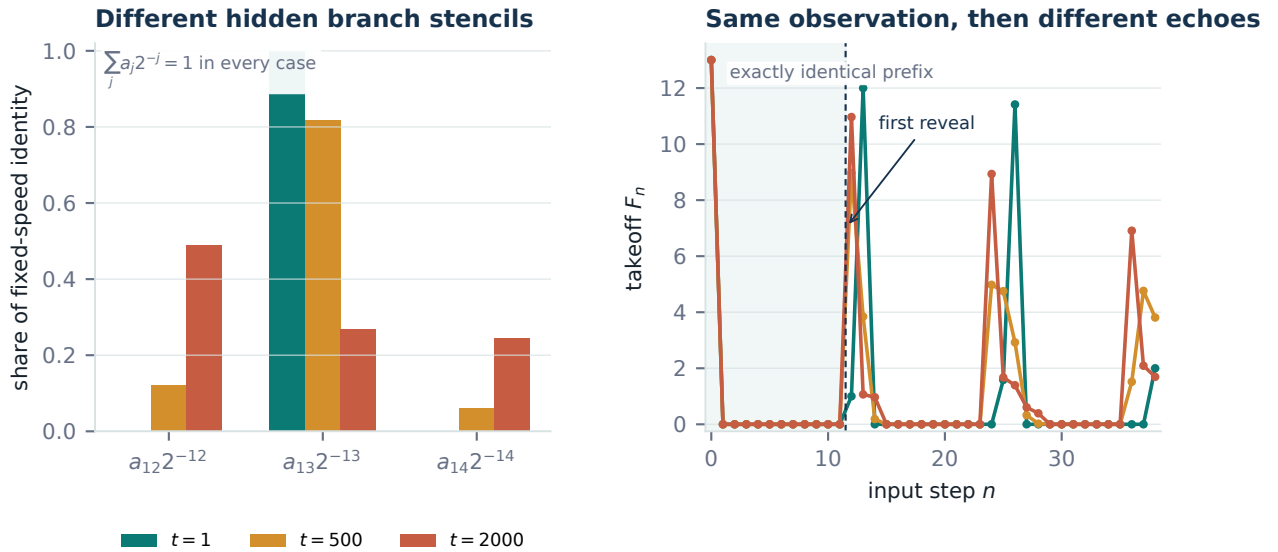


Figure 2: Exact finite-horizon blindness at $L = 12$. Left: three distinct coefficient stencils, plotted by their contribution $a_j 2^{-j}$ to the fixed-speed identity. Right: their takeoff profiles coincide exactly on $F_0, \dots, F_{11}$ (shaded) and first separate at F_{12} . Later pulse trains retain visibly different recursive signatures. This collision is algebraic, not a consequence of noise or an unsuccessful optimizer.

Table 1: Completed AWS runs.

Run	Schemas	Trials	Time
Broad baseline	575,440	13,810,560	18.6 min
Horizon sweep	589,302	18,857,664	14.2 min
Constraint boundary	38,796	2,327,760	12.2 min
Total	1,203,538	34,995,984	45.0 min

The baseline formed Hankel matrices

$$(H_0)_{ij} = \tilde{Y}_{i+j}, \quad (H_1)_{ij} = \tilde{Y}_{i+j+1},$$

where $i, j \geq 0$ range over entries whose subscripts lie in the observed window. It selected an order from one through six by held-out prediction and recovered poles from a matrix pencil truncated by singular-value decomposition (SVD). It then forecast 24 unseen samples. We scored the largest recovered pole modulus, qualitative class (monotone, alternating, or ringing), and holdout normalized root-mean-square error (NRMSE). The NRMSE is the holdout RMS prediction error divided by the maximum of the clean-window RMS, holdout RMS, and 10^{-12} . “Forecast success” means NRMSE below 0.1. Ground truth was used only after fitting. Leading-radius error is the absolute difference between the largest recovered and true normalized pole moduli.

Three independent seeded sweeps were run. The third compared the six-mode pencil with two literal source estimators: a nonnegative continuous fit followed by rounding, and an integer fit enforcing (2). Both source fits used the difference equations in Lemma 1. It tested $D \in \{8, 12, 16, 20\}$ at $T \in \{D-2, D-1, D, D+2, 2D+2\}$ with a 16-sample holdout. Table 1 reports compute time. At the verified rate of \$1.428/hour, estimated EC2 compute was \$1.10 total; the new run cost about \$0.32 and took 13.4 minutes wall time. All instances terminated and all temporary cloud resources were removed.

The broad run used lengths 24, 48, 96 and lags through 18. The horizon sweep used lengths 8, 16, 32, 64 and lags through 32, permitting explicit bins in T/D , where T is the window length and D is that schema’s maximum lag. Each schema contributed every window, regime, and noise condition; the schema generator itself was stochastic rather than an exhaustive uniform census.

5 Results

Figure 3 summarizes the two most important empirical boundaries: information arriving after the hidden lag, and prediction remaining reliable after modal identification degrades.

Sub-lag observations are blind.. Table 2 isolates noiseless finite-horizon traces beginning at the origin. Modal-class accuracy was zero throughout $T/D < 1$. This empirical zero is consistent with Corollary 7, although the exact result concerns

Table 2: Noiseless finite-horizon family, origin windows.

T/D	Trials	Class acc.	Forecast succ.
< 0.5	101,309	0.0%	4.9%
$[0.5, 1)$	132,303	0.0%	0.0%
$[1, 2)$	147,761	1.4%	0.0%
≥ 2	209,671	8.4%	10.9%

source recovery rather than a particular modal classifier’s error. There is no rapid recovery immediately after the lag: the generic six-mode baseline remained poor even for $T/D \geq 2$. These schemas can have degree 34 and crowded near-periodic modes, so the post-lag failure is an empirical limitation of the baseline, not a second impossibility theorem.

The estimator is not universally ineffective.. Random dyadic-split recurrences were much easier once observations began after the largest lag. In the noiseless broad run, class accuracy was 78.3%, 88.9%, and 86.6% for windows 24, 48, and 96; forecast success was 95.6%, 99.5%, and 99.8%. The corresponding sampled median leading-radius errors were 0.0024, 0.0007, and 0.0005. Thus the hard-family failure is structurally selective.

Prediction can survive a false explanation.. For post-lag random-split traces of length 96, raising relative noise from zero to 10^{-3} reduced modal-class accuracy from 86.6% to 32.1% and increased sampled median radius error from 0.0005 to 0.4453. Forecast success nevertheless remained 99.8%. A finite-window predictor can therefore extrapolate accurately while assigning the wrong modal geometry. This is the most practically important empirical warning: predictive fit, modal identification, and source identification are three different evidentiary levels.

Structural information closes the noiseless gap.. Figure 4 tests the positive side of Theorem 5. At $T = D - 2$, the integer optimizer matched the generating member in 45.0% of trials. That aggregate mixes uniquely determined boundary cases with arbitrary selection inside multi-schema version sets: 74.1% of schemas had more than one compatible tail, the median version count was 43, and the maximum was 174,763. One sample later it recovered all 38,796 noiseless sources; continuous fit-and-round recovered 2.3%, while the generic pencil does not output literal coefficients. Under relative noise 10^{-8} , 10^{-5} , 10^{-3} , the constrained exact-source rates were 99.98%, 70.1%, and 38.6%. Its corresponding forecast-success rates were 100%, 99.6%, and 94.2%, compared at 10^{-3} with 49.8% for continuous rounding and 19.7% for the pencil. Thus the algebraic threshold is exact in the noiseless model and becomes a graded statistical boundary under perturbation.

6 What the experiment establishes

The exact results characterize every finite-prefix equivalence class, exhibit exponentially many indistinguishable fixed-

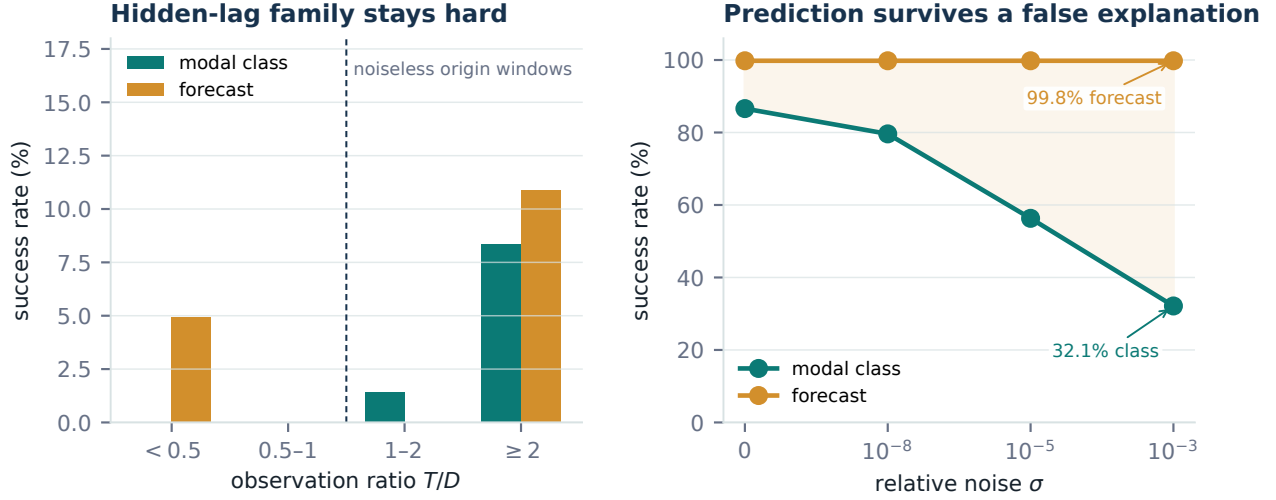


Figure 3: Recovery phase diagrams from the completed AWS sweeps. Left: on the constructed finite-horizon family, noiseless origin-window recovery is absent below the hidden lag and remains weak after it becomes visible. Right: on 144,863 post-lag random-split schemas with a 96-sample window, modal classification degrades sharply with relative noise while 24-step forecast success remains 99.8%. Bars and curves are computed directly from the saved run summaries.

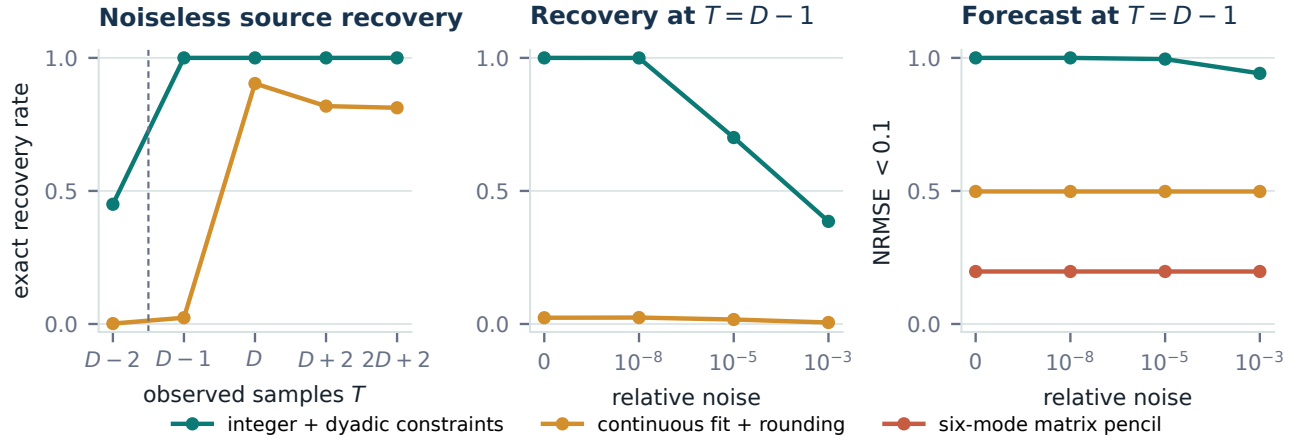


Figure 4: The structural constraints locate the sharp boundary. Left: in 38,796 noiseless schemas, the integer dyadic decoder jumps to 100% exact source recovery at the theorem’s $T = D - 1$ threshold; unconstrained fit-and-round reaches only 2.3% there. Center: exact recovery at the boundary under noise. Right: 16-step forecast success; the constrained source remains useful after exact coefficient recovery degrades. The $T = D - 2$ aggregate is not a uniform guarantee because the benchmark includes observationally equivalent source classes.

invariant sources, extend the obstruction to arbitrary parameter-independent noise, and give a constructive decoder once a lag bound is known. The experiment then shows where a standard spectral tool works and where it does not. Massive replication makes the family contrast stable; it does not turn the classical matrix-pencil algorithm into a novel method.

Several limitations matter. The schemas are synthetic and one-dimensional; real program traces may contain input-dependent branching and measurement artifacts. The modal baseline

is capped at order six, while the hardest recurrences have much larger degree. Root calculations for high-degree, near-periodic polynomials are themselves conditioned problems. The logarithmic takeoff profile is only asymptotically a finite exponential sum, and very small late corrections approach floating-point resolution. Finally, millions of trials do not remove bias from the chosen schema distribution.

The constraint comparison exploits information the baseline

ignores:

$$a_j \in \mathbb{Z}_{\geq 0}, \quad \sum_j a_j 2^{-j} = 1,$$

together with a declared lag bound. Its sharp noiseless transition confirms that the theorem is algorithmically attainable, while its noisy decline shows that identifiability does not imply good conditioning. Real recursive traces should ultimately test whether recovered equivalence classes remain stable outside the synthetic model.

7 Conclusion

The mathematical boundary is now explicit. A finite trace discloses its total branching and all coefficients below the observation horizon—no more and no less. Beyond that horizon, exponentially many fixed-speed, aperiodic sources can remain observationally identical; adding parameter-independent noise or a finite pre-return holdout cannot manufacture the missing information. On the positive side, a declared lag bound and the $D - 1$ exact samples through F_{D-2} recover the entire recurrence, and the construction proves that one fewer sample fails uniformly. Separately, common terminal speed does not control transient duration: near-periodic kernels can settle arbitrarily slowly.

The computation reinforces rather than replaces those results. Across 1.20 million hidden schemas and 35.0 million blinded trials, a classical matrix-pencil estimator performed well on many visible short-lag responses but failed selectively on the constructed hidden-lag family. With relative noise 10^{-3} , 24-step forecast success remained 99.8% while modal-class accuracy fell to 32.1%. At the sharp $D - 1$ boundary, the constrained decoder recovered 100% of noiseless sources and retained 94.2% forecast success at relative noise 10^{-3} . The central lesson is therefore a hierarchy of evidence: forecasting a response, identifying its modes, and recovering its recursive source are progressively stronger achievements. Claims about mechanism should stop exactly where the available horizon and structural assumptions stop.

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